

Ilari Angervuori

Wireless Systems Researcher (R&D) | System-Level Stochastic Analysis and Simulation | RF Link Modelling

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Profile

R&D scientist in wireless networks with a pioneering publication record in stochastic geometry modelling of non-terrestrial networks (NTN), including a first-author publication in the high-impact journal IEEE Transactions on Communications (IEEE TCOM) and a first-author presentation at the flagship conference IEEE GLOBECOM. Looking for a transition to industry. Enjoys applying existing knowledge in novel environments. Committed to bringing my unique expertise to a collaborative, development-driven, fast-paced environment at a company that is intertwined with emerging AI tools. A mindset that seeks simplicity – not complexity. Always striving to grow professionally and spiritually, and to cultivate compassion and humility in all aspects of life. Available to relocate; FI/EU citizen.

Skills

Soft:

• Data analysis and visualisation • Inter-institute collaboration • Scientific writing • Research • Stakeholder management

Hard:

• 3GPP NR specifications, particularly NTN • Communication theory • Physical Layer and MAC simulation and analysis • Statistical Digital Signal Processing • System-level performance analysis of wireless networks • MATLAB, JAVA, C, C++

Spoken Languages:

• Finnish (native) • English (fluent) • Swedish (limited working proficiency) • Spanish (Basics, no working proficiency)

Experience and Impact

5/2025- **Visiting Research Scientist**, Aalto University, School of Electrical Engineering, Department of Information and Communications Engineering, Espoo, Finland

- Managed 2 R&D projects on mathematical modelling of wireless communications as the corresponding author, and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication.
- Put forth a closed-form autocorrelation function for aggregate radio frequency (RF) Physical Layer interference at a narrow-beam low Earth orbit (LEO) satellite terminal. Leveraged tractable spectrum analysis and hypothesis classes for machine learning and artificial intelligence (ML/AI) signal processing algorithms, with applications in interference control and medium access control (MAC) design in satellite communications.
- Proposed and analysed successive interference cancellation (SIC) algorithms for multibeam LEO uplink, improving coverage performance by 900% for the third-strongest Earth transmitter in terms of signal-to-interference ratio (SIR). The SIC improves user fairness and enables serving multiple terrestrial terminals per cell, while preserving the average total network throughput at the LEO satellites, which is optimised around 1 (bit/s/Hz). As a trade-off, throughput slightly decreases compared to a one-user-per-cell link. The mathematical analysis was tested and validated against more realistic Monte Carlo simulations. The system-level insights imply design recommendations for next-generation LEO networks.
- Peer-reviewed journals (including top-impact), IEEE TCOM, IEEE TWC, IEEE TMC, etc.
- Mentored junior researchers as a supervisor and helped establish future talents in wireless communications.

8/2023- **Collaborative Research Fellowship**, University of Notre Dame, Department of Electrical Engineering,
2/2024 South Bend, IN, US

- Secured project financing from the Väisälä Foundation for an RF communications R&D project in collaboration with the world's top experts in wireless networks research and development.
- Coordinated an internationally distributed team and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication in the high-impact journal IEEE TCOM.
- Developed a novel mathematical framework for stochastic analysis of LEO networks, leveraging tractable, strikingly simple, closed-form expressions for various performance metrics, crucial in optimal network configuration design and more efficient allocation of simulation resources.
- Built a novel system model and mathematical analysis of link budget and Physical Layer RF interference in a LEO uplink, characterising coverage probability (SIR distribution) using a Lomax distribution. The analysis was tested and validated against more realistic Monte Carlo simulations. Consequently, Lomax distribution results can be utilised for considerably more efficient performance analysis and optimisation of LEO communications, providing insight and serving as a hypothesis class in ML/AI algorithms. The results were published in a high-impact journal, IEEE TCOM.
- Identified an optimal cell size in a LEO uplink that maximises average throughput and quantified a trade-off between average throughput and performance consistency: -10 dB cell size provides high average throughput but highly variable link quality of service (QoS) across the beams, whereas -3 dB cells provide more stable QoS but are detrimental to throughput. The results were published in a high-impact journal, IEEE TCOM.

- 09/2020- **PhD Researcher**, *Aalto University*, School of Electrical Engineering, Espoo, Finland
- 05/2025
- Built, maintained, and documented a multi-language simulation stack (GIT, MATLAB, MATHEMATICA, OCTAVE, PYTHON) including algorithms for analysis, simulation, and visualisation, used across several research projects.
 - Developed a novel, well-cited stochastic geometry framework for a LEO RF uplink that reduced complexity and simulation time compared to MATLAB Satellite Communication Toolbox full-fidelity simulations while producing similar coverage/throughput performance results (statistically relevant results were obtained from initialisation to computation in minutes rather than hours). The project was a collaboration with the Technical Research Centre of Finland (VTT).
 - Put forth model simplifications that allow the LEO uplink Physical Layer RF interference to be characterised by a gamma distribution. The model was tested and validated against more realistic Monte Carlo simulations. Consequently, leveraged well-known gamma distribution properties for efficient performance analysis and optimisation, insight, and as a hypothesis class for ML/AI algorithms. Presented the paper at the top-tier conference IEEE GLOBECOM in Rio De Janeiro.
 - Proposed, analysed, and simulated a minimal-overhead inter-satellite link (ISL) data-routing algorithm in large constellations using graph algorithms. Optimised a grant-free MAC to maximise throughput and minimise latency.
 - Gained an outstanding number of citations (> 300 and counting), implying high-impact research on stochastic modelling of LEO networks.

Education

- Expected
2026
- Doctor of Science, PhD**, *Aalto University, School of Electrical Engineering*, Signal processing for communications, • Stochastic geometry analysis and simulation of satellite communications
- MSc**, *Univ. of Helsinki*, Applied analysis, • Finite element method (FEM) • Helmholtz equation
- BSc**, *Univ. of Helsinki*, Applied analysis, • Optimal control theory • System performance optimisation